

Model Lab

Every prior, same slate — 2026-09-15

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

35,357

Held-Out Scores

Fitted on plate appearances before 2026-08-10, scored on the 35,357 that came after (actual HR rate 3.00%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5652	1.34×	0.02907	0.13407	0.00446
KNN comps Prior is the pooled rate of the nearest comparables.	0.5821	1.47×	0.02906	0.13384	0.00519
SVR Support-vector regression on the same feature matrix.	0.5706	1.34×	0.02917	0.13587	0.00923
Random forest Forest regression; a learned similarity metric.	0.5742	1.36×	0.02913	0.13522	0.00846
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5821	1.47×	0.02906	0.13384	0.00519
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5813	1.47×	0.02907	0.13392	0.00522
Linear regression Ordinary least squares on the same features.	0.5777	1.28×	0.02911	0.13478	0.00781
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5799	1.40×	0.02911	0.13443	0.00767
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5786	1.28×	0.02910	0.13437	0.00823
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5721	1.29×	0.02919	0.13564	0.00986

Blend weights (forest variant): KNN 0.90, SVR 0.00, RF 0.10

Shrinkage K per prior: league 245, knn 245, svr 245, rf 245, core 245, ensemble 245, linear 245, logistic 245, nn 245, xgb 245 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,738 PA, actual HR rate 2.46%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5030	0.98×	0.02404	0.11605	0.01012
KNN comps Prior is the pooled rate of the nearest comparables.	0.5600	0.98×	0.02413	0.11741	0.01467
SVR Support-vector regression on the same feature matrix.	0.5342	0.87×	0.02472	0.12955	0.02735
Random forest Forest regression; a learned similarity metric.	0.5427	0.87×	0.02453	0.12586	0.02418
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5600	0.98×	0.02413	0.11741	0.01467
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5580	0.98×	0.02416	0.11790	0.01471
Linear regression Ordinary least squares on the same features.	0.5533	0.98×	0.02443	0.12441	0.02135
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5627	0.98×	0.02430	0.12028	0.02108
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5608	1.09×	0.02429	0.11991	0.01879
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5471	0.87×	0.02498	0.12838	0.03152

Slate Agreement

Flat league · 5/6 shared 2/6 hit

Junior Caminero ✓
Pete Crow-Armstrong
Jackson Merrill
Kyle Schwarber ✓
Victor Mesa Jr.
Max Muncy

SVR · 5/6 shared 2/6 hit

Junior Caminero ✓
Kyle Schwarber ✓
Shea Langeliers
Pete Crow-Armstrong
Max Muncy
Jackson Merrill

Core (KNN+SVR) · 6/6 shared 2/6 hit

Lazaro Montes
Jackson Merrill
Kyle Schwarber ✓
Junior Caminero ✓
Max Muncy
Pete Crow-Armstrong

Linear regression · 5/6 shared 2/6 hit

Lazaro Montes
Emmanuel Rodriguez
Kyle Schwarber ✓
Jackson Merrill
Junior Caminero ✓
Max Muncy

Neural net (torch) · 5/6 shared 2/6 hit

KNN comps · 6/6 shared 2/6 hit

17.3%	Lazaro Montes	19.3%
16.3%	Jackson Merrill	18.6%
16.0%	Kyle Schwarber ✓	18.4%
15.8%	Junior Caminero ✓	18.3%
15.4%	Max Muncy	18.1%
15.1%	Pete Crow-Armstrong	17.9%

Random forest · 6/6 shared 2/6 hit

20.6%	Lazaro Montes	21.6%
20.0%	Jackson Merrill	19.2%
19.3%	Kyle Schwarber ✓	19.0%
18.9%	Junior Caminero ✓	18.7%
18.8%	Pete Crow-Armstrong	18.4%
18.6%	Max Muncy	18.0%

Forest blend · 6/6 shared 2/6 hit

19.3%	Lazaro Montes	19.5%
18.6%	Jackson Merrill	18.7%
18.4%	Kyle Schwarber ✓	18.5%
18.3%	Junior Caminero ✓	18.3%
18.1%	Max Muncy	18.1%
17.9%	Pete Crow-Armstrong	18.0%

Logistic regression · 4/6 shared 1/6 hit

24.3%	Lazaro Montes	28.8%
19.5%	Emmanuel Rodriguez	23.6%
19.3%	Kyle Schwarber ✓	21.7%
18.7%	Jackson Merrill	19.7%
18.6%	Max Muncy	19.4%
18.0%	Shea Langeliers	19.3%

Pete Crow-Armstrong

Emmanuel Rodriguez

18.4%

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Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Josue De Paula Los Angeles Dodgers · vs RHP · 1 HR / 14 PA	14.2%	17.3%	4.8%	18.0%	17.3%	17.4%	16.4%	11.2%	14.0%
9. Shea Langeliers Athletics · vs RHP · 23 HR / 420 PA	14.3%	17.1%	19.3%	18.0%	17.1%	17.2%	17.6%	19.3%	18.7%
10. Pete Alonso Baltimore Orioles · vs RHP · 35 HR / 656 PA	14.7%	17.0%	17.6%	17.5%	17.0%	17.1%	17.0%	17.6%	17.2%
11. Yohandy Morales Washington Nationals · vs LHP · 2 HR / 27 PA	11.9%	16.4%	17.0%	17.1%	16.4%	16.4%	16.5%	16.4%	17.3%
12. Willson Contreras Boston Red Sox · vs LHP · 26 HR / 522 PA	12.7%	16.3%	15.5%	16.0%	16.3%	16.3%	16.2%	17.9%	16.4%
13. Corey Seager Texas Rangers · vs RHP · 18 HR / 385 PA	13.1%	16.3%	16.6%	16.4%	16.3%	16.3%	16.2%	16.5%	16.7%
14. Matt Olson slate Atlanta Braves · vs LHP · 39 HR / 654 PA	13.9%	16.2%	17.4%	16.8%	16.2%	16.3%	16.2%	17.4%	16.9%
15. Elly De La Cruz Cincinnati Reds · vs RHP · 27 HR / 585 PA	14.8%	16.1%	17.3%	15.9%	16.1%	16.0%	15.9%	16.4%	16.3%
16. Zack Gelof Athletics · vs RHP · 18 HR / 386 PA	14.5%	16.0%	16.4%	16.2%	16.0%	16.0%	16.3%	16.2%	16.4%
17. Ronald Acuña Jr. Atlanta Braves · vs LHP · 17 HR / 431 PA	14.4%	15.9%	15.9%	15.6%	15.9%	15.9%	15.4%	15.6%	15.8%
18. Coby Mayo Baltimore Orioles · vs RHP · 23 HR / 411 PA	13.3%	15.9%	17.2%	16.6%	15.9%	16.0%	16.2%	17.9%	16.8%
19. Miguel Vargas Chicago White Sox · vs LHP · 31 HR / 653 PA	12.9%	15.8%	15.6%	15.1%	15.8%	15.7%	15.8%	16.4%	16.0%
20. Kyle Stowers Miami Marlins · vs RHP · 19 HR / 477 PA	13.0%	15.7%	15.8%	15.0%	15.7%	15.7%	14.7%	15.7%	15.6%
21. Carter Jensen Kansas City Royals · vs RHP · 24 HR / 558 PA	13.8%	15.7%	15.8%	15.4%	15.7%	15.7%	14.8%	15.5%	15.6%
22. Yordan Alvarez	13.2%	15.7%	16.9%	16.4%	15.7%	15.8%	16.1%	17.5%	16.7%

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Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Riley Greene Detroit Tigers · vs RHP · 20 HR / 545 PA	12.7%	15.0%	14.8%	14.9%	15.0%	14.9%	14.3%	15.0%	14.9%
38. Justin Foscue Texas Rangers · vs RHP · 11 HR / 229 PA	12.3%	14.8%	15.0%	14.0%	14.8%	14.7%	14.2%	14.2%	14.6%
39. Andrés Chaparro Washington Nationals · vs LHP · 10 HR / 202 PA	13.4%	14.8%	14.9%	14.6%	14.8%	14.7%	14.2%	14.2%	14.8%
40. Lawrence Butler Athletics · vs RHP · 16 HR / 488 PA	14.5%	14.6%	14.6%	14.4%	14.6%	14.6%	14.8%	14.4%	14.5%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.